

A Study of the Onsite H-BIM Prototyping of Korean Traditional Wooden Architecture

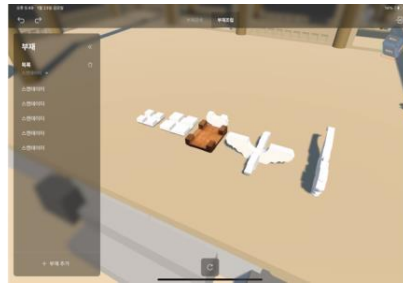
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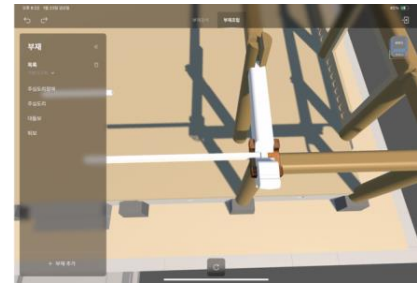


Figure 1: Development of an Onsite HBIM Prototyping System for Korean Traditional Wooden Architecture: Interpretation and Virtual Assembly (left: prototyping start screen, center: onsite 3D scanning and placement of template components, right: virtual assembly process)

ABSTRACT

Historic Building Information Modeling (H-BIM) offers clear benefits for the digital preservation of architectural heritage. In Korea, many traditional wooden buildings still lack H-BIM data due to the complexity and labor required for component-level modeling. In this paper, we propose an onsite mobile system for H-BIM prototyping with 3D scan of architectural components, virtual assembly and direct input of preservation information. It enables rapid generation of component-based H-BIM data in the field. It is expected to be a way to provide 3D preservation management capabilities for heritage buildings without H-BIM data.

Index terms: H-BIM, Onsite, Prototyping, VR, Virtual assembly, Korean traditional wooden architecture

1 INTRODUCTION

Digitizing architectural heritage has become a mainstream approach to preservation. Various digital documentation methods are used for the documentation and preservation of cultural heritage, including 3D scan and H-BIM [6].

3D scan is very useful for digitally recording the current shape of a historic building [1]. As it captures realistic surface geometry, it is widely used to creation of cultural heritage content too. However, it has limitations for the

heritage preservation workflows that require the component-level representation and information. In the process of preservation, architectural components must be created and their information managed individually. 3D scan often produces a chunk of point clouds or meshes that represents the exterior shape [2]. Since they are not divided into architectural components, historic building's internal structure or preservation-related components information could not be represented. Therefore, additional tools and works are needed.

H-BIM extends Building Information Modeling (BIM) technology used in modern buildings to historic architectures. BIM is designed for managing building information throughout its lifecycle. While BIM for modern buildings focuses on design and construction of new building, H-BIM starts from onsite survey and measuring of existing historic buildings and carries out preservation activities, like visualization, status information recording, and analyses [5]. H-BIM fundamentally manages buildings at the component level and includes preservation information. H-BIM can be easily used for digital integration for preservation.

Although H-BIM has many advantages, H-BIM data for all architectural heritage sites is not available. Generating H-BIM data requires specialized knowledge and a lot of work, depending on the scale of the building [3]. To create and store preservation information for buildings without H-BIM data, it is necessary to prototype H-BIM data rapidly. Since architectural heritage sites are outdoors, it is preferable to prototyping by mobile devices, referencing actual buildings.

We propose a mobile system to prototype H-BIM data for Korean traditional wooden architecture at a heritage site. It supports measuring the components or parts of buildings by mobile 3D scan, virtually assembling them, directly entering preservation information, and finally generating H-BIM data.

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Direct onsite mobile prototyping can support an intuitive and efficient method for creating 3D component-based H-BIM data. The preservation information is primarily collected through onsite surveys and stored in a connected database.

In Korea, only about 110 wooden heritage buildings have H-BIM data [7]. For architectural heritage sites lacking H-BIM data, creating onsite virtual assemblies of building components coupled with preservation information can support scalable H-BIM prototyping and preservation documentation for heritage sites. Therefore, this paper's contribution is as follows.

- An onsite scan-to-HBIM prototyping method using traditional wooden components
- Fast generation of highly reliable H-BIM data through virtual assembly that verifies the actual building
- A preservation pipeline with onsite H-BIM data and the survey information at heritage site

The following effects are expected through this paper's method.

- Opportunities to make H-BIM data for more architectural heritage sites
- Higher shape quality in H-BIM data with a combination of 3D scan and conventional H-BIM
- Digital storage of preservation information based on 3D H-BIM

2 CHARACTERISTICS OF TRADITIONAL WOODEN ARCHITECTURE

Korean traditional wooden structures are composed of wooden components that are fitted and joined using traditional joinery techniques. Each member is carefully shaped to ensure a tight fit at the joints. As a result, a single building typically consists of numerous members, and the roof structural system depicted in Figure 2 often exhibits particularly intricate configurations.

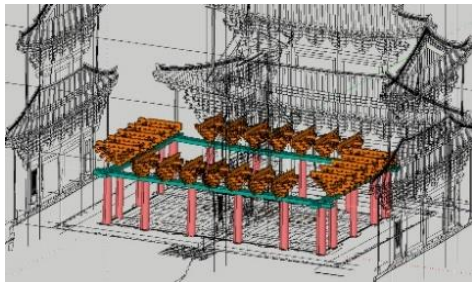


Figure 2: Example of Korean traditional wooden architecture and its internal structure

2.1 H-BIM for Korean Traditional Wooden Architecture

In some H-BIM workflows for Western architectural heritage, which often involves stone or masonry structures, practitioners rely on 3D scan data to capture overall

geometry for documentation. In contrast, 3D scans of traditional Korean wooden architecture are typically used to capture exterior geometry as a point cloud and serve as a digital record of the current state.

H-BIM modeling for traditional Korean wooden structures often emphasizes component-level representation due to the complexity of the structural system and joinery [9]. Based on onsite measurements, 3D data for individual components are created. The building hierarchy is then constructed according to architectural conventions, and attribute information is prepared for each component [4]. Preservation-related records are entered at the component level, as shown in Figure 3.

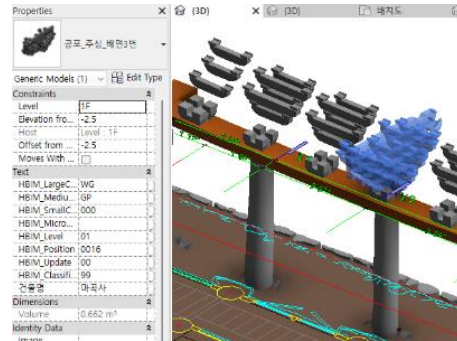


Figure 3: Example of H-BIM data of Korean traditional wooden architecture

3 ONSITE PROTOTYPING

This section describes an onsite prototyping method for rapidly generating simplified H-BIM data by virtually assembling architectural components captured in the field. Based on the characteristics of traditional Korean wooden architecture and the H-BIM requirements discussed in the previous section, we implemented three key functions: (1) mobile 3D scanning of components or building parts, (2) Virtual assembly supported by joint-aware placement between components, (3) A user interface for entering preservation metadata for each component.

The overall prototyping process consists of four stages, the first three of which are carried out onsite. These are illustrated in Figure 4 and described below.

4. Creating the 3D shape of an architectural component

We provide two methods to obtain 3D data for building components or parts. One is a mobile 3D scan, the other is a template-based component.

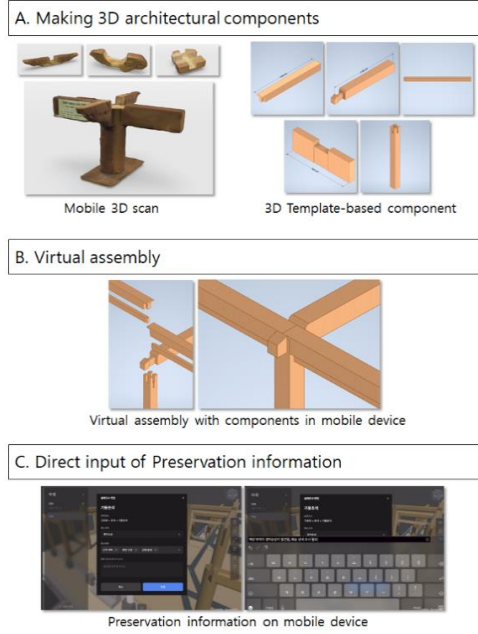


Figure 4: Onsite prototyping process at the heritage building.

1) Mobile 3D scanning: 3D data is captured using a mobile scanning application on an iPad Pro equipped with a LiDAR sensor. Figure 5 shows the scanning process, and Figure 6 shows the reconstruction results. The left side presents the captured point cloud and mesh, while the right side shows the texture-mapped mesh [1].



Figure 5: 3D scanning using an iPad Pro



Figure 6: 3D scanning results

2) Template-based component: We also prepared a library of 3D templates for components commonly used in traditional Korean wooden architecture and archived them for reuse. Users can inspect the target structure onsite and select the required components from a list. Figure 7 shows the interface for selecting and loading a template component.

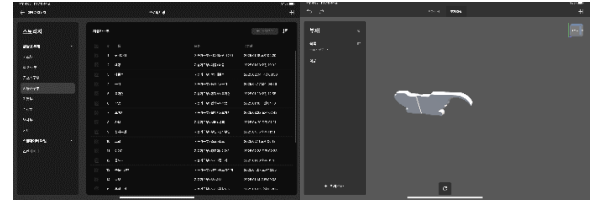


Figure 7: Selecting and loading architectural component template

B. Virtual Assembly

In this step, the user assembles a structure on a mobile device using previously acquired 3D components. We provide an assembly and editing interface within a VR workspace.

To enable intuitive object manipulation, we implemented standard 3D transformation tools for translation, rotation, and scaling. We also provide a dedicated panel for editing component class labels such as columns and beams.

In addition to transformation, the system supports duplication and deletion to facilitate rapid onsite prototyping.

After completing H-BIM prototype, users can select each component. To enable joint-aware placement, the system detects the surface of the existing component that the user selects when placing a new component. The snapping function estimates the pose of the target surface and places the selected component so that the contact surfaces align without visible gaps. This improves placement consistency during the prototyping step.

We also implemented part-based visualization and filtering so that users can display only selected categories of components during assembly based on their classification criteria.

C. Direct input of Preservation Information

As described in the previous section, users can select each component and enter preservation-related metadata. This interface supports recording the current condition and onsite survey results at the component level and saving them for later review and updates.

4 EXPERIMENTS AND RESULTS

This section describes experiments on virtual assembly for H-BIM prototyping using three approaches. The experiments include prototyping using 3D scan data, prototyping using template-based 3D component data, and a hybrid approach that combines both.

The experiments were conducted as follows. After completing the virtual assembly and generating a prototype model, preservation metadata was entered for each component using the same workflow intended for H-BIM implementation.

4.1 Prototyping using 3D scan

Because architectural heritage structures cannot be physically disassembled for testing except during dismantling and repair, 3D scanning and virtual assembly

were performed using a scaled model.

4.2 Prototyping using 3D template components

We tested template-based prototyping using Nangnamheon in Hwaseong, Suwon, a UNESCO World Heritage Site. Nangnamheon is a hall used for official events and banquets, built in 1794. Together with the adjacent Noraedang Hall, it is among the few buildings at the site that have not been reconstructed in modern times [8].

4.3 Hybrid Prototyping using 3D scan and template

In this study, prototyping was performed by combining traditional architectural component templates with 3D scan data. As shown in Figure 8, preservation metadata for each component was entered using the same workflow as in the H-BIM prototyping results for Nangnamheon.



Figure 8: Final result of H-BIM Prototyping of Nangnamheon in Hwaseong

5 CONCLUSIONS

In this paper, we propose an onsite H-BIM prototyping method combining mobile 3D scan and virtual assembly of components from Korean traditional wooden structures. This approach targets architectural heritage sites lacking pre-existing H-BIM data and enables the rapid creation of component-based models in the field.

We also introduced a consistent prototyping workflow that integrates standard templates with 3D scans. Our workflow provides an intuitive interface for component-level assembly and inspection. By enabling onsite linkage between H-BIM components and survey metadata, the approach supports three-dimensional preservation documentation for partial structures and facilitates subsequent refinement and updates.

Although generating full-building H-BIM remains challenging under onsite constraints, we establish a practical framework for producing partial H-BIM data and managing preservation information in a 3D environment. Beyond conventional mobile content applications, the approach suggests a direction for using mobile spatial computing to support cultural heritage preservation workflows.

In the future, we will focus on reducing manual effort by automating key onsite and post-processing steps. We will explore AI-assisted recognition to accelerate scan cleanup and component identification. Moreover, we aim to extend the approach by integrating XR-based tele-preservation,

enabling offsite access to inspection and preservation workflows beyond onsite constraints. In addition, we plan to collaborate with relevant organizations to increase H-BIM coverage and improve interoperability with Korea's national H-BIM initiatives.

ACKNOWLEDGMENTS

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